CS & IT ENGINEERING

Computer Organization Architecture

Cache Organization

Discussion Notes

DPP- 02

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[MCQ]

#Q. Consider a 512KB direct mapped cache with block size of 32 bytes. The main memory address is of 34-bits. The size of index and tag in bits are?

2^5 B

A. 15, 14

B. 14, 15

C. 19, 14

D. 14, 19

Tag	block	byte
15	14	5

34

$\log 512K = 19$

no. of blocks in cache $= \frac{512KB}{32B} = \frac{2^{19}}{2^5} = 2^{14}$

[NAT]

Ans = 64

#Q. Consider a direct mapped cache of size 256MB. Cache controller maintains 8-bits tag for each block in cache. The maximum size of main memory (byte addressable) supported in the system is _____GB?

Tag	block	byte

8 ⟷ $\log 256M = 28$

mm add. = 36 bits

mm size = 2^{36} bytes

= 64 GB

[NAT]

Ans = 256

#Q. The size of memory required at cache controller to store metadata is 2KBytes. The metadata includes tag bits, 1 modified bit and 1 valid bit. The cache contains 1K blocks of 16 bytes each and organized as direct mapped. The size of main memory is___ Mbytes?

mm add.

Tag	Block	byte
14	10	4

mm add. = 28 bits

mm size = 2^{28} bytes

= 256 MB

Tag directory size = no. of blocks * (Tag+2) bits

$2\cancel{K} * 8$ bits = $1\cancel{K} * (Tag + 2)$

Tag = 14 bits

[NAT]

Ans = 32

#Q. Consider a direct mapped cache of size 256KB. The CPU generates x- bit addresses. The number of tag bits in main memory address are 14 bits then value of x is _____?

mm add.

Tag	cm block	byte

14

$\longleftrightarrow$ log 256k = 18 bits

mm add. = 14 + 18

= 32 bits

[MCQ]

#Q. Assume a computer has 32-bit addresses. Each block stores 64 bytes. A direct-mapped cache has 512 blocks. Match the block (line) of the cache (in decimal) we look for each of the given hexadecimal addresses in the table?

A. 1A2BC012: 256, FFFF00FF: 3, 12345678: 345, C109D532: 340

B. 1A2BC012: 512, FFFF00FF: 7, 12345678: 243, C109D532: 320

C. 1A2BC012: 128, FFFF00FF: 5, 12345678: 345, C109D532: 420

D. 1A2BC012: 255, FFFF00FF: 1, 12345678: 247 , C109D532: 240

← 32 →		
Tag	block	byte
17	9	6

1.) 1A2BC012

	9 bits	
$\boxed{1A2B}_{16}$ 1	100 000000	010010
	$\Downarrow$ $(256)_{10}$	

2) $\boxed{FFFF}$00FF $(0000000011)_2 = (3)_{10}$

3) $(\boxed{1234}5678)_{16}$ $(101011001)_2 = (345)_{10}$

4) $(C109D532)_{16}$ $(101010100)_2 = (340)_{10}$

[NAT]

Ans = 4

2^5

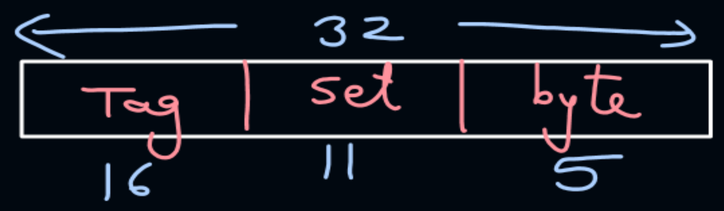
#Q. Consider a cache with 2^{13} blocks of size 32 Bytes each. The CPU generates addresses of 32-bits. The cache controller stores 1 valid bit, 1 modified bit and tag-bits for each metadata entry. The cache controller has a maximum memory of 18Kbytes to store the metadata. The cache is organized as k-way set associative. Maximum value of k to utilize the cache controller memory in optimized manner is ___?

Tag directory size = 18 KB = $2^{13} * (\text{Tag} + 2)$ bits

$18\cancel{K} * \cancel{8}$ bits $= \cancel{2^{13}} * (\text{Tag} + 2)$ bits

Tag = 16 bits

mm add.



$$\text{no. of sets} = 2^{11} = \frac{\text{no. of blocks in cm}}{k}$$

$$2^{11} = \frac{2^{13}}{k}$$

$$k = \frac{2^{13}}{2^{11}} = 2^{2} = \underline{\underline{4}}$$

[NAT]

Ans = 97.8 %

#Q. Consider a direct mapped write back data cache of size 2KB with the block size of 128 bytes. The cache is considered to be empty initially. The byte addressable main memory has size 1Mbytes. Further consider that there is an array A[35][20] with each element occupies 4 bytes. The base address of array is $(1A300)_{16}$. The array is accessed 3 times. And between the accesses, there is no any data cache changes happen. Hit ratio (correct upto 1 decimal place) of cache for this array access is ___%?

no. of blocks in cache $= \frac{2kB}{128B} = \frac{2^{11}}{2^{7}} = 2^{4} = 16$

Array size $= 35 * 20 = 700$ elements

$= 700 * 4 = 2800$ Bytes

no. of blocks needed to store array $= \left\lceil \frac{2800\,B}{128\,B} \right\rceil = 22$ blocks

overlapping blocks $= 22 - 16 = 6$

	no. of miss
1st access	22
2nd access	$6 * 2 = 12$
3rd "	$6 * 2 = 12$
Total	46

Total no. of mem. reference $= 700 * 3$
$= 2100$

No. of hits $= 2100 - 46 = 2054$

Hit ratio $= \frac{2054}{2100} = 0.978 = 97.8\%$

[NAT]

#Q. Consider a direct mapped cache of size 32 Kbytes. The cache uses 64 bytes block. Consider main memory address is of 20 bits. For the main memory address $(94347)_{10}$, the tag is $(___)_{10}$?

No of blocks in cache $= \frac{32\ KB}{64\ B} = 512$

Block no. of the given memory addr $= \left\lfloor \frac{94347}{64} \right\rfloor = 1474$

Tag $= \frac{1474}{512} = 2$

[NAT]

#Q. Consider a ~~2~~ 4-way set associative cache of size 256 Kbytes. The cache uses 32 bytes block. Consider main memory address is of 20 bits. For the main memory address $(86147)_{10}$, the tag is (____)~~10~~ 10?

No of blocks in cache $= \frac{256\ KB}{32\ B} = 8192$

No of sets in cache $= \frac{8192}{4} = 2048$

Main mem block no $= \frac{86147}{32} = 2692$

Tag $= \left\lfloor \frac{2692}{2048} \right\rfloor = 1$

[NAT]

#Q. Consider a fully associative cache of size 16Kbytes. The cache uses 128 bytes block. Consider main memory address is of 20 bits. For the main memory address $(5182)_{10}$, the tag is (____)$1\cancel{0}_{10}$?

Main memory block no = Tag value = $\frac{5182}{128}$ = 40

[MCQ]

No of sets : $\frac{8}{2} = 4$

#Q. A 2-way set associative cache with LRU cache replacement contains 8 blocks. The CPU requests for main memory blocks in following sequence:

d d d d d t x t x t x d d d x d d x x x x y y

8, 12, 0, 1, 5, 8, 1, 12, 8, 0, 1, 3, 6, 7, 3, 11, 2, 3, 6, 2, 8, 7, 11

Number of cold, capacity and conflict misses respectively are?

0	0	8
1	1	5
2	6	2
3	11	7

$8 \% 4 = 0$

$0 \% 4 = 0$

$1 \% 4 = 1$

$5 \% 4 = 1$

$3 \% 4 = 3$

$6 \% 4 = 2$

$7 \% 4 = 3$

$11 \% 4 = 3$

$2 \% 4 = 2$

A 10, 3, 2

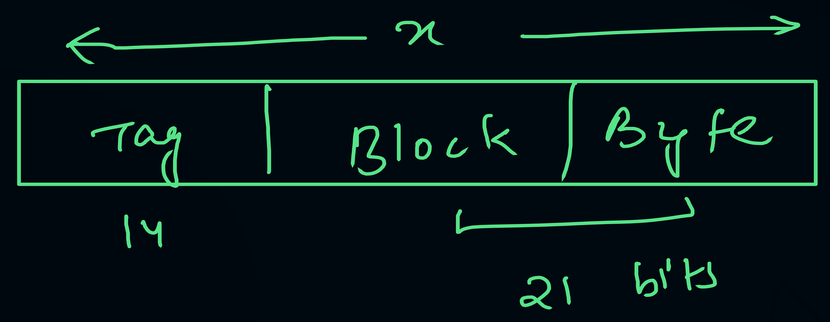
B 10, 2, 3

C 9, 3, 2

D 9, 2, 3

[NAT]

#Q. Consider a direct mapped cache of size 2MBytes. The CPU generates x-bit addresses. The number of tag bits in main memory address are 14 bits then value of x is ___?



Cache size = 2MB

$= 2^{21}$ B

21 bits

$x = 14 + 21 = 35$ bits

[NAT]

#Q. Consider a set-associative cache of size 16KB (1KB=2^{10} bytes) with cache block size of 32 bytes. Assume that the cache is byte-addressable and a 32-bit address is used for accessing the cache. If the width of the tag field is 20 bits, the associativity of the cache is _____?

32		
Tag	Set	Byte
20	7	5

Set $= 32 - 20 - 5 = 7$

No of blocks $= \dfrac{2^{14}\ B}{2^{5}\ B} = 2^{9}$

No. of sets $= 2^{7}$

Associativity $= \dfrac{2^{9}}{2^{7}} = 4$

[NAT]

#Q. Assume a computer has 32 bits addresses. Each block stores 128 bytes. A 4-way set associative cache has size 512Kbytes. The set of the cache would we look for the address $(AC1016AF)_{16}$ is (_____)$_{10}$?

32

Tag	Set	Byte
15	10	7

X

No of blocks $= \frac{512\ KB}{128\ B} = 2^{12}$

No of sets $= \frac{2^{12}}{4} = 2^{10}$

0 0001 0110 1

1 8 1 $= (45)_{10}$

THANK - YOU